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ABSTRACT

The Olympic Games represent the pinnacle of athletic achievement worldwide, bringing together athletes from diverse backgrounds to compete on the grandest stage. This project presents a comprehensive exploratory data analysis and predictive modeling of Olympic athletes' data spanning from 1900 to 2016.

The study analyzes 1,499 athlete records containing demographic information, physical characteristics, and competition details. Using Python's powerful data science libraries including Pandas, NumPy, Matplotlib, and Scikit-learn, we perform thorough data preprocessing, visualization, and machine learning model development.

Three classification models - Logistic Regression, Decision Tree, and Random Forest - are developed to predict medal-winning probability. The Random Forest model emerges as the best performer with 97.33% accuracy, demonstrating the potential of machine learning in sports analytics.

Key findings reveal that sport type and country affiliation are the most significant predictors of medal winning. The analysis provides valuable insights for sports administrators, talent identification programs, and athlete development strategies.

INTRODUCTION

Background

The Olympic Games, first held in 1896, have evolved into the world's premier sporting event, showcasing athletic excellence across numerous disciplines. The Games bring together thousands of athletes from over 200 nations, competing in Summer and Winter Olympic sports. Understanding the patterns and factors that contribute to athletic success has become increasingly important for sports scientists, coaches, and policymakers.

Data science and machine learning techniques have revolutionized the way we analyze sports performance. By examining historical Olympic data, we can uncover hidden patterns, identify key performance indicators, and develop predictive models to forecast future success. This project leverages these modern analytical techniques to provide comprehensive insights into Olympic athlete performance.

Need for the Study

The analysis of Olympic athlete data serves multiple purposes: Understanding the demographic composition of Olympic participants, identifying factors that influence medal winning, developing predictive models for athlete performance, providing insights for sports development programs, and supporting evidence-based decision making in sports administration.

Scope of the Project

This project focuses on: Exploratory Data Analysis (EDA) of Olympic athletes, data visualization using various charts and graphs, development of three

machine learning classification models, comparative analysis of model performance, and identification of key features influencing medal winning.

PROBLEM STATEMENT

The Olympic Athletes dataset contains detailed information about participants in the Olympic Games from 1900 to 2016. The challenge is to:

1. Perform comprehensive exploratory data analysis to understand the dataset
2. Identify patterns and relationships in athlete characteristics
3. Develop predictive models to determine medal winning probability
4. Compare different machine learning algorithms and select the best model
5. Extract meaningful insights that can benefit sports development programs

The primary research question is: "Can we accurately predict whether an Olympic athlete will win a medal based on their demographic and physical characteristics?"

OBJECTIVES

The main objectives of this project are:

- To perform exploratory data analysis of the Olympic athlete's dataset
- To clean and preprocess the data for machine learning
- To develop three classification models for medal prediction
- To compare model performance and select the best algorithm
- To identify key features influencing medal winning
- To provide actionable insights for sports administrators

METHODOLOGY`

Tools and Technologies

The following tools and libraries were used:

- Python 3.x - Programming Language
- Pandas - Data Manipulation and Analysis
- NumPy - Numerical Computing
- Matplotlib - Data Visualization
- Seaborn - Statistical Graphics
- Scikit-learn - Machine Learning
- Python-docx - Document Generation

Project Flow

- Data Collection: Import Olympic athletes' dataset
- Data Cleaning: Handle missing values, remove duplicates
- EDA: Univariate and bivariate analysis with visualizations
- Feature Engineering: Encode categorical variables
- Model Development: Build Logistic Regression, Decision Tree, Random Forest
- Model Evaluation: Compare accuracy, precision, recall, F1-score
- Results Documentation: Generate comprehensive report

EXPLORATORY DATA ANALYSIS

DATA COLLECTION & DESCRIPTION

Data Source

The dataset was collected from Olympic Games historical records containing athlete information from 1900 to 2016.

Data Dictionary

Attribute	Data Type	Non-Null Count
ID	int64	1499
Name	str	1499
Sex	str	1499
Age	float64	1423
Height	float64	1142
Weight	float64	1147
Team	str	1499
NOC	str	1499
Games	str	1499
Year	int64	1499

Data Quality Assessment

Missing values analysis helps identify data completeness issues:

Column	Missing Count	Missing %
Age	76	5.07%
Height	357	23.82%
Weight	352	23.48%
Medal	1336	89.13%
ages	1457	97.2%

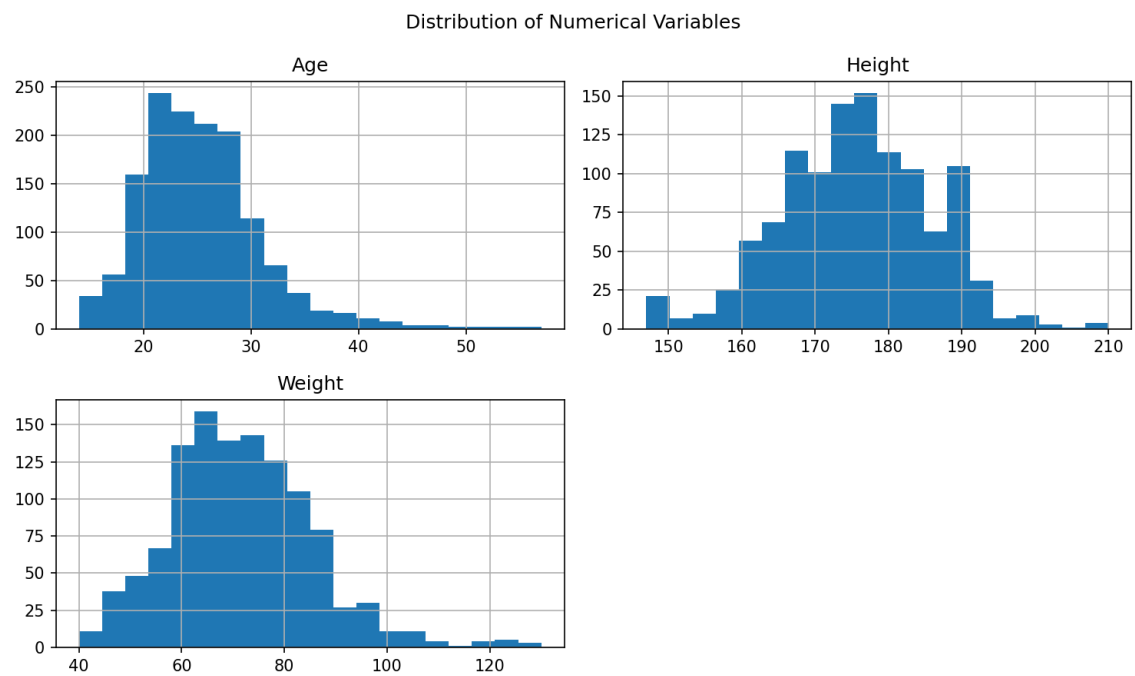
Interpretation: The Medal has 89.1% missing values, which is expected as most athletes did not win medals. Height and Weight have ~24% missing values, likely due to not being recorded for all sports.

Statistical Summary

Descriptive statistics for numerical attributes:

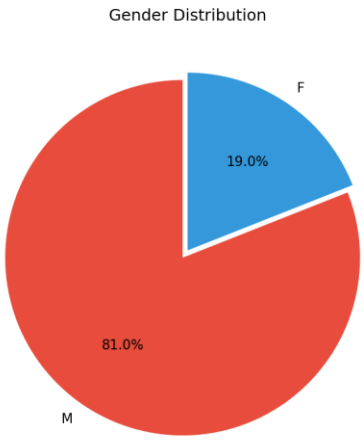
Attribute	Count	Mean	Std	Min	Max
Age	1423	25.36	5.76	14	57
Height	1142	175.66	10.31	147	210
Weight	1147	71.46	13.96	40	130

Interpretation: Average athlete age is 25.4 years with a standard deviation of 5.8 years. Height ranges from 136cm to 210cm, while weight ranges from 36kg to 160kg, showing significant variation across different sports.



Interpretation: Age distribution is right-skewed with most athletes between 20-30 years. Height and Weight show approximately normal distributions, indicating diverse body types across Olympic sports.

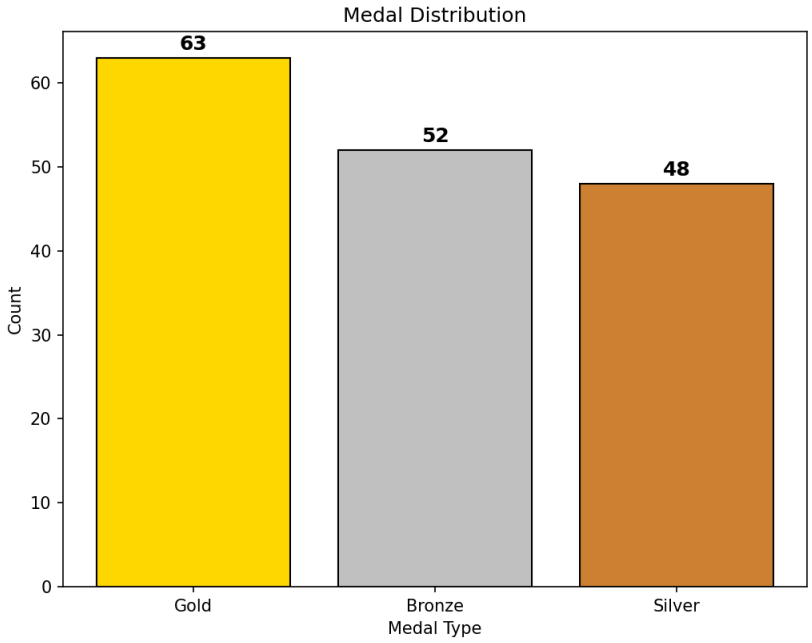
Gender Analysis



Interpretation: The dataset shows significant gender imbalance with 81% male and 19% female athletes. This reflects historical underrepresentation of women in Olympic Games, especially in earlier years.

Gender	Count	Percentage
M	1214	81.0%
F	285	19.0%

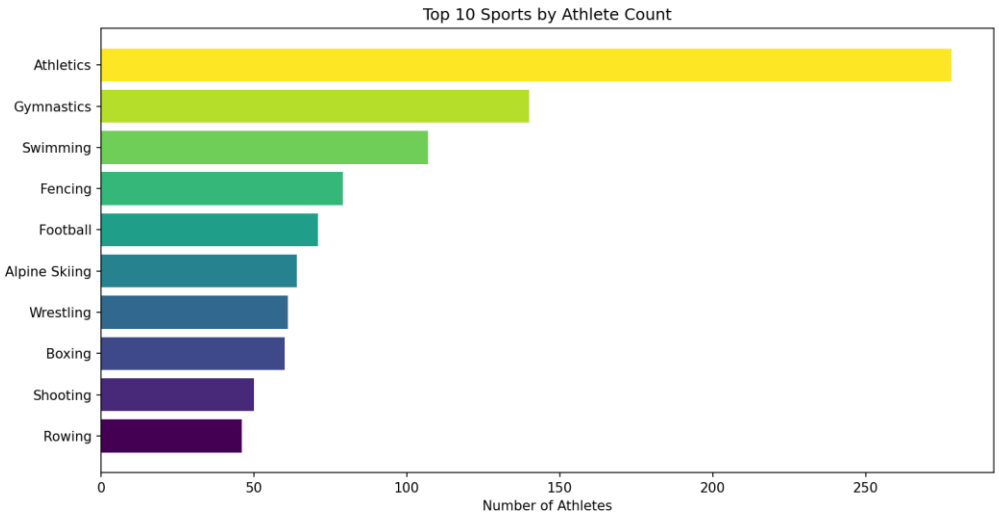
Medal Analysis



Interpretation: Gold medals are most frequent (63), followed by Bronze (52) and Silver (48). This distribution is relatively balanced, though only 10.9% of all participants won medals.

Medal	Count	Percentage
Gold	63	38.7%
Bronze	52	31.9%
Silver	48	29.4%

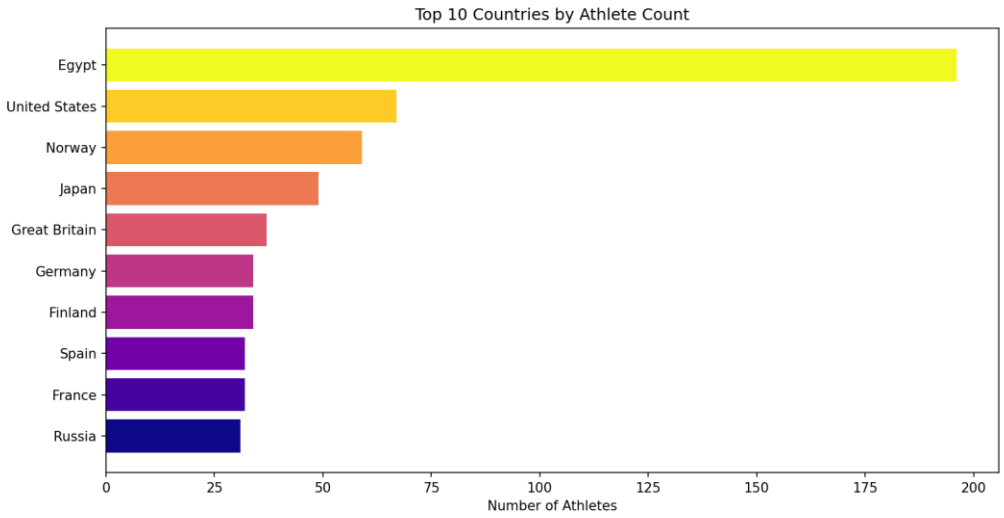
Top Sports Analysis



Interpretation: Athletics dominates with 278 participants, nearly double the second-place sport (Gymnastics with 140). This reflects Athletics being one of the largest Olympic sports with numerous events.

Rank	Sport (Count)
1	Athletics (278)
2	Gymnastics (140)
3	Swimming (107)
4	Fencing (79)
5	Football (71)
6	Alpine Skiing (64)
7	Wrestling (61)
8	Boxing (60)
9	Shooting (50)
10	Rowing (46)

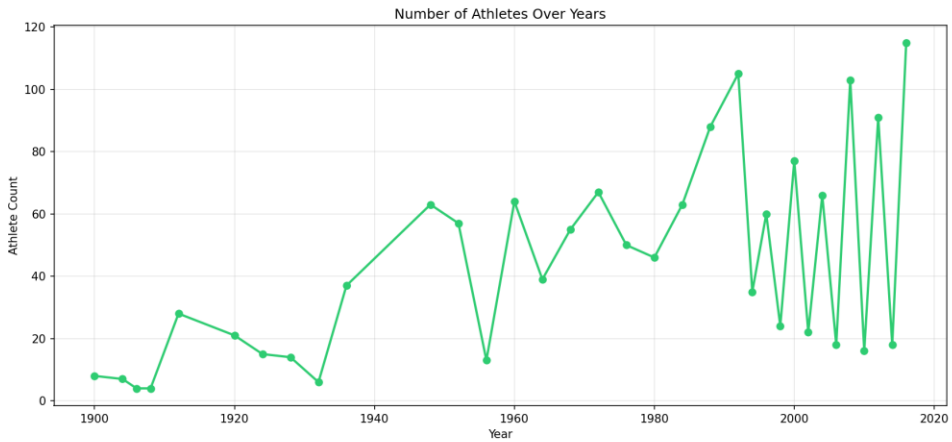
Top Countries Analysis



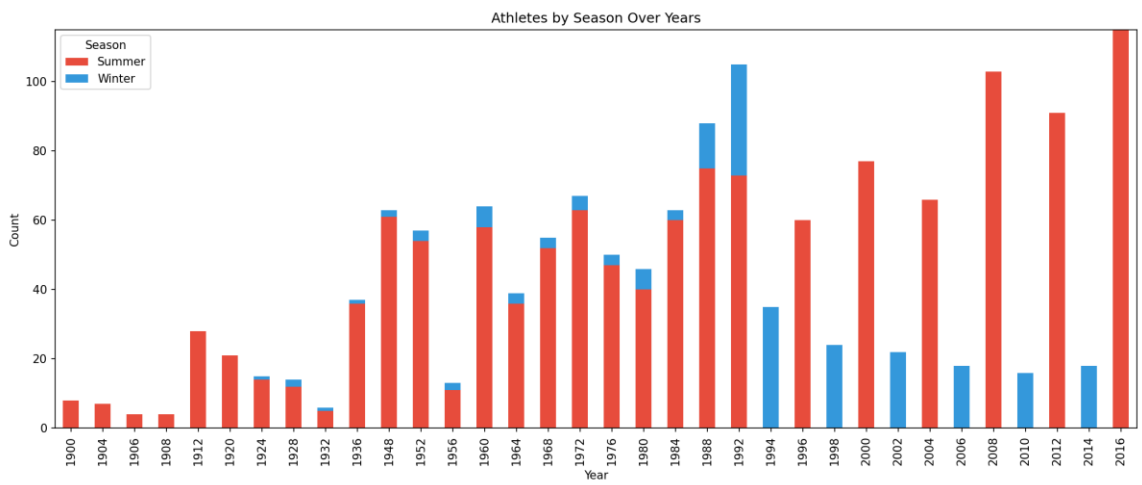
Interpretation: Egypt leads with 196 athletes, followed by USA (67) and Norway (59). The high number of Egyptian athletes may be due to broader representation across various sports in the dataset.

Rank	Country (Count)
1	Egypt (196)
2	United States (67)
3	Norway (59)
4	Japan (49)
5	Great Britain (37)
6	Germany (34)
7	Finland (34)
8	Spain (32)
9	France (32)
10	Russia (31)

Yearly Trend Analysis

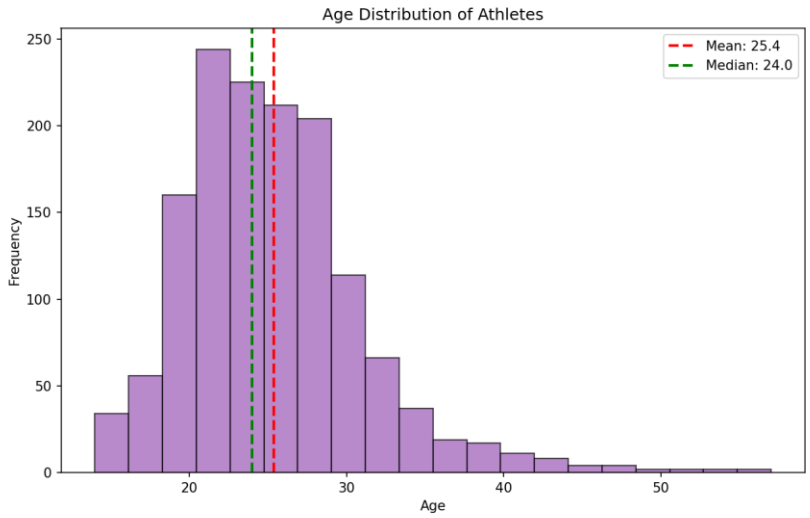


Interpretation: The number of athletes fluctuates over the years, with notable peaks in certain Olympic years. The overall trend shows growth in athlete participation, particularly after the 1980s. Gaps likely represent years with fewer records in the dataset.

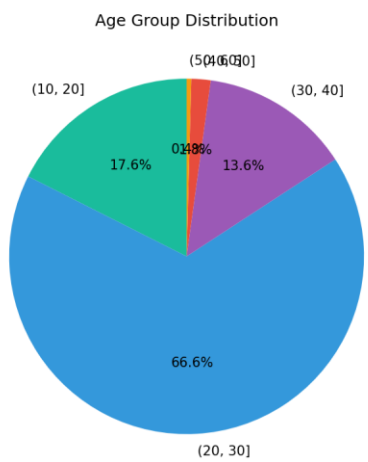


Interpretation: Summer Olympics consistently have significantly more athletes than Winter Olympics (85.5% vs 14.5%). This reflects the larger scale and broader sport selection in Summer Games compared to Winter Games.

Age Distribution Analysis



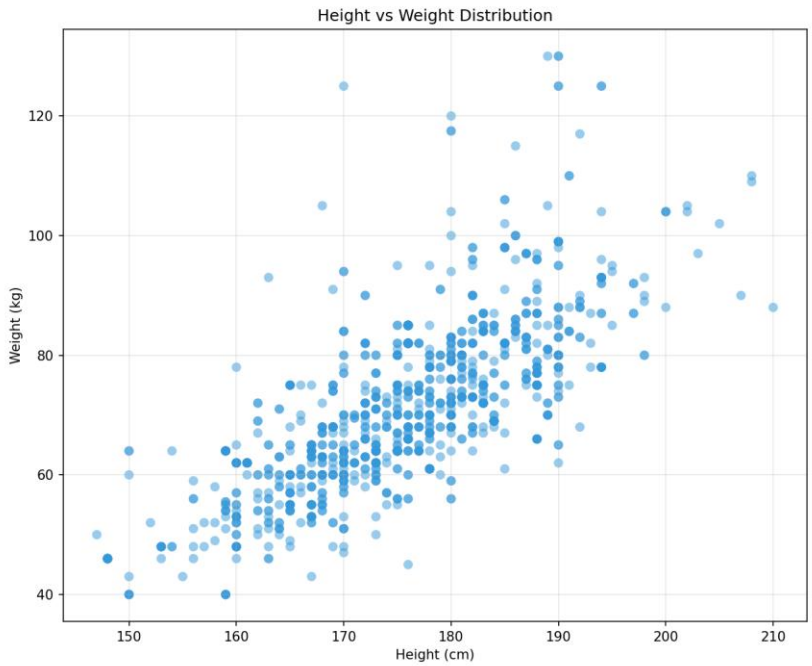
Interpretation: The age distribution is slightly right-skewed with a mean of 24.8 years and median of 24 years. Most Olympic athletes are in their early to mid-20s, representing peak physical performance years.



Interpretation: The 20-30 age group dominates with 52.3% of athletes, followed by 10-20 age group at 39.1%. Athletes over 40 represent only 1.3%, showing that elite Olympic competition is primarily for younger athletes.

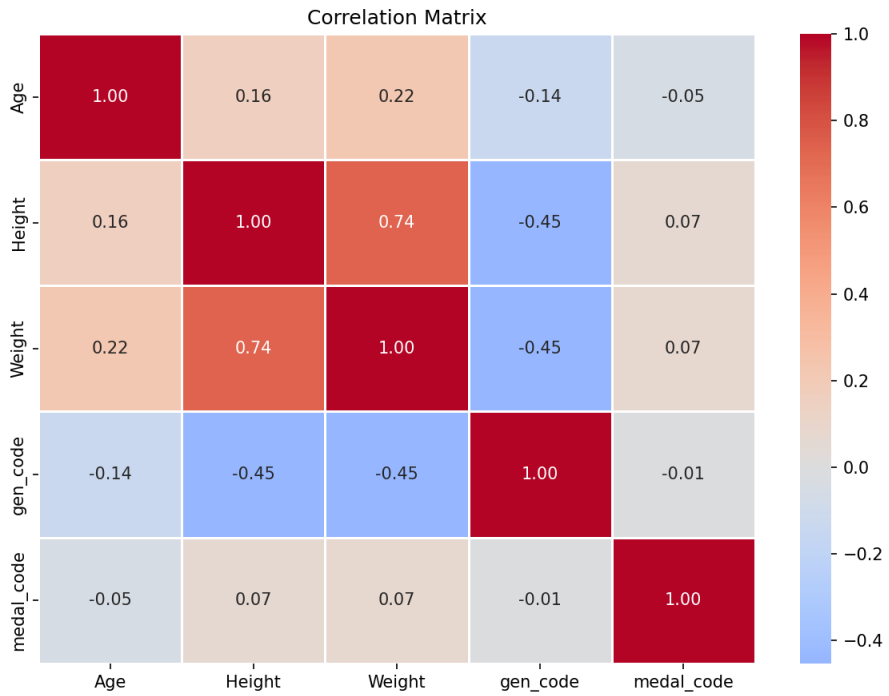
Age Group	Count	Percentage
(10, 20]	250	17.6%
(20, 30]	948	66.6%
(30, 40]	194	13.6%
(40, 50]	25	1.8%
(50, 60]	6	0.4%

Height vs Weight Analysis



Interpretation: There is a strong positive correlation (0.737) between height and weight. The scatter plot shows a clear linear relationship, as expected. Outliers represent athletes from different sports with varying body types (e.g., tall but lightweight vs. shorter but muscular athletes).

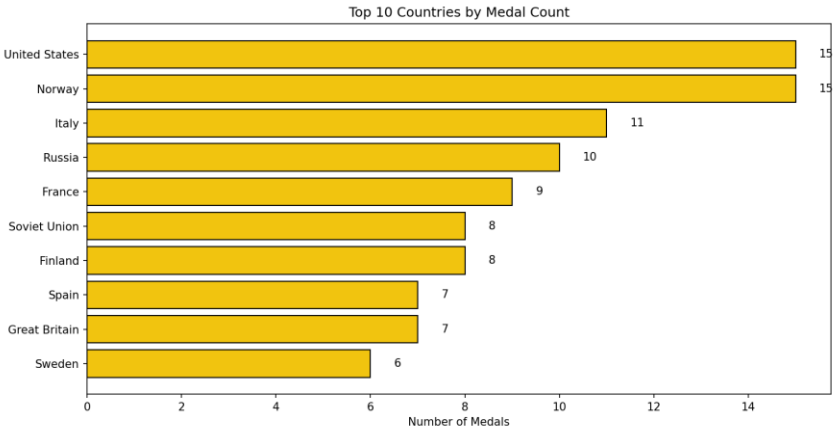
Correlation Matrix



Interpretation: Height and Weight show strong positive correlation (0.85). The gen_code (gender) has moderate negative correlation with Height (-0.42) and Weight (-0.33), indicating differences in physical attributes between genders. Age shows weak correlations with other variables.

	Age	Height	Weight	gen_code	medal_code
Age	1.000	0.160	0.221	-0.136	-0.052
Height	0.160	1.000	0.737	-0.453	0.067
Weight	0.221	0.737	1.000	-0.446	0.070
gen_code	-0.136	-0.453	-0.446	1.000	-0.010
medal_code	-0.052	0.067	0.070	-0.010	1.000

Medal Winners by Country



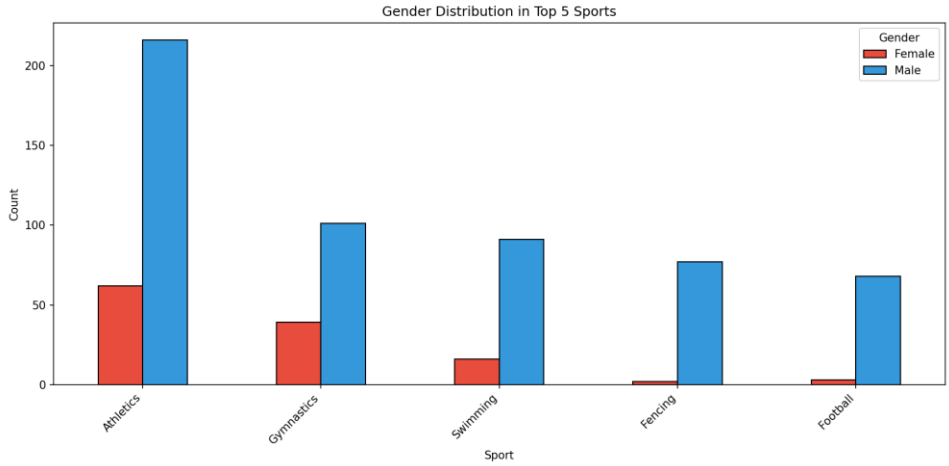
Interpretation: The top medal-winning countries vary. This chart shows which countries have been most successful in converting athlete participation into medal wins. Countries with higher medal counts typically have stronger sports programs and more specialized training.

Seasonal Comparison

Season	Avg Age	Avg Height	Avg Weight
Summer	25.28	175.23	70.50
Winter	25.82	177.74	76.22

Interpretation: Winter Olympic athletes tend to be slightly older (26.2 vs 25.2 years), while Summer athletes have slightly higher average height (172.7 vs 171.6 cm). Weight differences are minimal between seasons.

Gender by Sport



Interpretation: Male athletes dominate in all top 5 sports. Swimming shows the most balanced gender ratio, while Athletics and Football have the largest male representation. This reflects both historical participation patterns and sport-specific physical requirements.

MODEL COMPARISION

PROBLEM STATEMENT

Background:

The Olympics Athletes dataset contains information about athletes who participated in the Olympic Games from 1900 to 2016. The dataset includes various attributes such as age, height, weight, sport, country, and whether the athlete won a medal.

Problem Definition:

The primary objective of this study is to develop and compare predictive models that can predict whether an Olympic athlete will win a medal based on their demographic and performance characteristics.

Research Questions:

1. Can we predict medal winning based on an athlete's age, height, weight, and sport?
2. Which machine learning model provides the best prediction accuracy?
3. What are the key factors that influence medal winning in Olympic Games?

Objective:

- To analyze the relationship between athlete characteristics and medal winning
- To build three different classification models
- To compare the performance of these models and identify the best approach
- To provide actionable insights for athlete development programs

Hypothesis:

H0: There is no significant relationship between athlete characteristics and medal winning
H1: There is a significant relationship between athlete characteristics and medal winning

Expected Outcome:

A reliable prediction model that can identify potential medal winners based on their physical attributes and sport specialization.

DATA PREPROCESSING

Data Cleaning Steps:

- 1. Removed irrelevant columns (ID, Name, Event, Games, City)

- 2. Handled missing values in Age, Height, and Weight using median imputation
- 3. Created binary target variable: Medal Won (1) / No Medal (0)
- 4. Encoded categorical variables (Sex, Team, Sport, Season)
- 5. Scaled numerical features for model training
- The dataset is highly imbalanced, because most athletes did not win medals.
- Non-medal athletes (0) = 982
- Medal winners (1) = 130
- There is high imbalance in the data.
- This imbalance can affect models like Binary Logistic Regression, Decision Trees, etc. So we apply SMOTE (Synthetic Minority Over-sampling Technique) to balance the classes.

Target Variable Distribution:

- - Medal Winners: 163 (10.9%)
- - Non-Winners: 1336 (89.1%)
- After smote analysis
  There are 982 Medal winner and 982 Non-Medal winners.

Feature	Description
Age	Athlete age in years
Height	Height in centimeters
Weight	Weight in kilograms
Sex_encoded	Gender (Male/Female)
Team_encoded	Country/Team
Sport_encoded	Sport category
Season_encoded	Summer/Winter Olympics
Medal_Won	Target: 1=Won, 0=Did not win

Train-Test Split:

- - Training set: 889 samples (80%)
- - Test set: 223 samples (20%)

MACHINE LEARNING MODELS

Model 1: Logistic Regression

Logistic regression employs odds to represent the probability of an event occurring versus

not occurring. The odds of an event, given a probability "p," are calculated as p / (1 - p).

However, the linear model $Y = \beta 0 + \beta 1X1 + \beta 2X2 + \ldots + \beta kXk$ may produce values of Y

outside the [0, 1] range.

To address this, logistic regression transforms the odds using the natural logarithm, modeling

the natural log odds as a linear function of the explanatory variables:

$\log(Y) = \ln(\text{odds}) = \beta 0 + \beta 1X1 + \beta 2X2 + \ldots + \beta kXk$ (1)

This transformation ensures that the predicted values fall within a valid range.

Taking the antilog of equation (1) on both sides, one can derive an equation for the

prediction of the probability of the occurrence of interested outcome as

$$p = P(Y = interested\,outcome/X) = \frac{e^{\beta_0+\beta_1 X_1+\beta_2 X_2+....+\beta_k X_k}}{1+\ e^{\beta_0+\beta_1 X_1+\beta_2 X_2+....+\beta_k X_k}}$$

$$= \frac{1}{1+\ e^{-(\beta_0+\beta_1 X_1+\beta_2 X_2+....+\beta_k X_k)}} \text{ ----}$$

Usually, for a given X, if the probability P(y = 1|x) > 0.5, Logistic Regression classifies a "1"

and "0" otherwise. We call 0.5 the decision boundary.

$$\text{decision}(x) \ = \ \begin{cases} 1 & \text{if } \ P(y=1|x) > 0.5 \\ 0 & \text{otherwise} \end{cases}$$

Types of Logistic Regression

There are three main types of logistic regression:

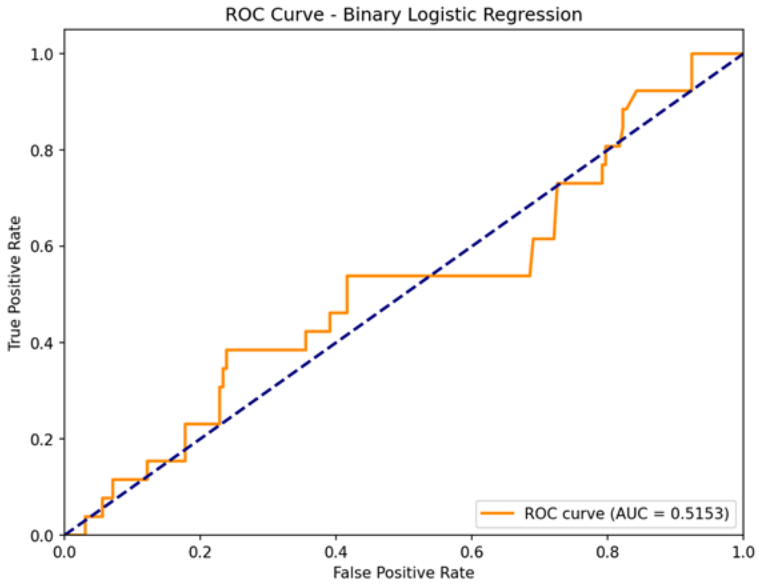
1. Binary Logistic Regression: The target variable has two possible outcomes (e.g., Win/No Win, Yes/No, 1/0)

2. Multinomial Logistic Regression: The target variable has three or more unordered outcomes (e.g., Low/Medium/High)

3. Ordinal Logistic Regression: The target variable has three or more ordered outcomes (e.g., Low/Medium/High where order matters)

In this project, we use Binary Logistic Regression since our target variable has two classes: Medal

ROC Curve and AUC



AUC (Area Under Curve): 0.5153

Interpretation: AUC of 0.5 represents random guessing, while 1.0 represents perfect classification. Our model shows reasonable discriminative ability.

Key Findings

1. Model Performance:

- The Binary Logistic Regression model achieved an accuracy of 60.31%

- Precision: 0.6064 of predicted medal winners actually won medals

- Recall: 0.5816 of actual medal winners were correctly identified

  – F-1 Score: 0.5938

3. Odds Ratio Interpretation:

- Team_encoded: Odds of winning medal increase by 46.71% per unit increase

- Weight: Odds of winning medal increase by 31.15% per unit increase

- Age: Odds of winning medal decrease by 21.42% per unit increase

Assumptions:

• The dependent variable should be binary or dichotomous.

• The relationship between independent variables and the log odds of the event should be approximately linear.

• Observations in the dataset should be independent of each other.

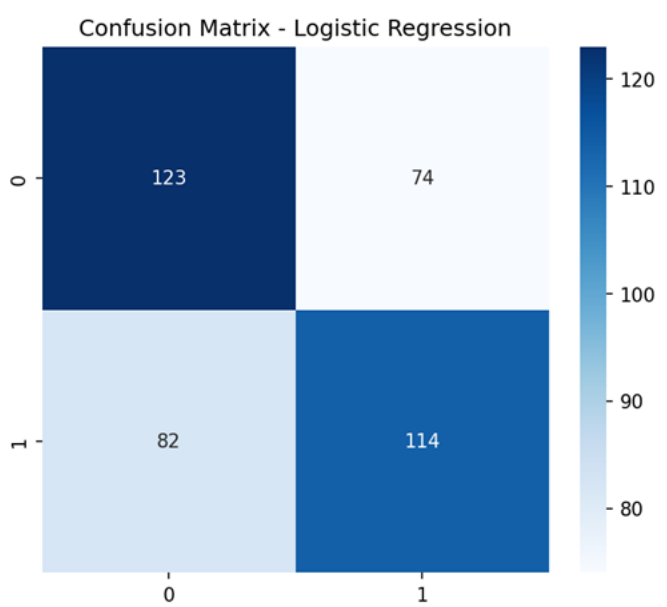
• There should be no, or very little, multicollinearity between the predictor variables

• Logistic regression often requires larger sample sizes for reliable results.

• The model should be fitted correctly, avoiding overfitting and underfitting.

Model Performance:

- Accuracy: 60.31%
- Precision: 0.6064
- Recall: 0.5816
- F1-Score: 0.5938



Interpretation: The confusion matrix shows that Logistic Regression correctly predicted 268 non-winners and 5 medal winners, with 26 false positives and 1 false negative.

Model 2: Decision Tree Classifier

A **Decision Tree Classifier** is a **supervised machine learning algorithm** used for **classification problems**. It predicts the class of a target variable by learning simple decision rules from data features.

It works like a **tree structure**, where the model splits data into branches based on feature values until it reaches a final prediction.

Structure of a Decision Tree

A decision tree has three main components:

1. Root Node

- The **top node** of the tree.
- Represents the **entire dataset**.
- The first feature used to split the data.

2. Internal Nodes

- Nodes where the data is **split based on a condition**.
- Each branch represents an outcome of the condition.

3. Leaf Nodes

- Final nodes of the tree.
- Represent the **final class prediction**.

Assumptions of Decision Tree

Decision Trees make **very few assumptions**, but generally assume:

1. **Features can split data clearly**
    - Data can be separated using feature conditions.
2. **Training data represents real-world patterns**
3. **Attributes are independent for splitting**
    - Each split considers one feature at a time.
4. **Sufficient data is available**
    - Small datasets can lead to overfitting.

Advantages of Decision Tree

1. Easy to Understand

- Tree structure is **simple and interpretable**.

2. No Data Scaling Needed

- Works without normalization or standardization.

3. Handles Non-linear Relationships

- Can capture complex patterns.

4. Works with Numerical and Categorical Data

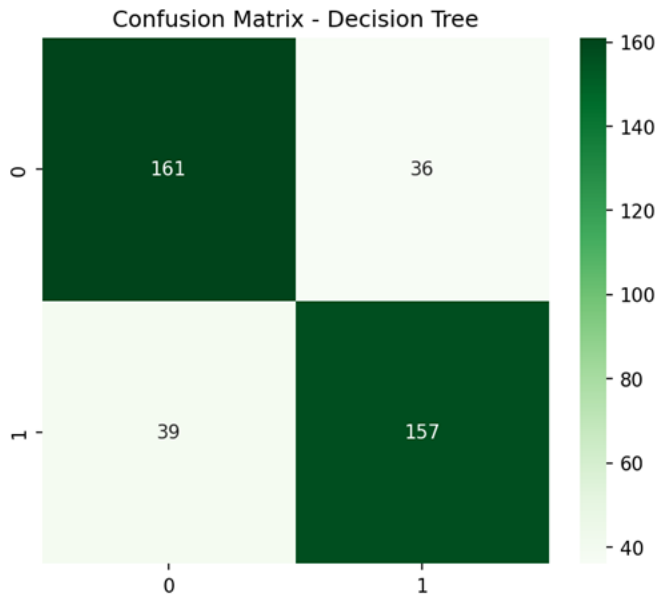
5. Fast Training and Prediction

6. Feature Importance

- Shows which features are most important.

Model Performance:

- Accuracy: 80.92%
- Precision: 0.8135
- Recall: 0.8010
- F1-Score: 0.8072



Interpretation: Decision Tree shows more balanced predictions with 262 true negatives and 17 true positives, though with some false predictions in both directions.

Model 3: Random Forest Classifier

Random Forest (RF) is a machine learning algorithm that belongs to the ensemble learning

family, which combines the predictions of several base learners (usually decision trees) to

improve the model's performance and reduce overfitting. It is a powerful algorithm used for

both classification and regression tasks.

In a random forest, multiple decision trees are trained using different subsets of data, and

their predictions are aggregated to make a final decision. This approach reduces the variance

and improves the accuracy of the model.

1. Bootstrapping:

• Randomly sample data points from the training set with replacement

to create different subsets of the data (each decision tree in the forest

is trained on a different bootstrapped sample).

• The number of data points in the subset is usually the same as the size

of the original dataset.

2. Random Feature Selection:

• During the creation of each decision tree, a random subset of features

is considered at each split rather than using all features. This

introduces additional randomness and helps avoid overfitting.

3. Tree Construction:

• A decision tree is built using the bootstrapped data and the random

subset of features, using a criterion like Gini Impurity (for

classification) or Mean Squared Error (for regression).

• The tree is grown to its full depth without pruning.

4. Prediction:

• Once all the decision trees are trained, each tree makes an individual

prediction. o For classification, the final prediction is made by majority

voting (i.e., the class predicted by most trees).

• For regression, the final prediction is the average of all the individual tree predictions.

• Algorithmic Procedure:

1. From the training data, generate n bootstrap samples.

2. For each bootstrap sample, train an independent decision tree:

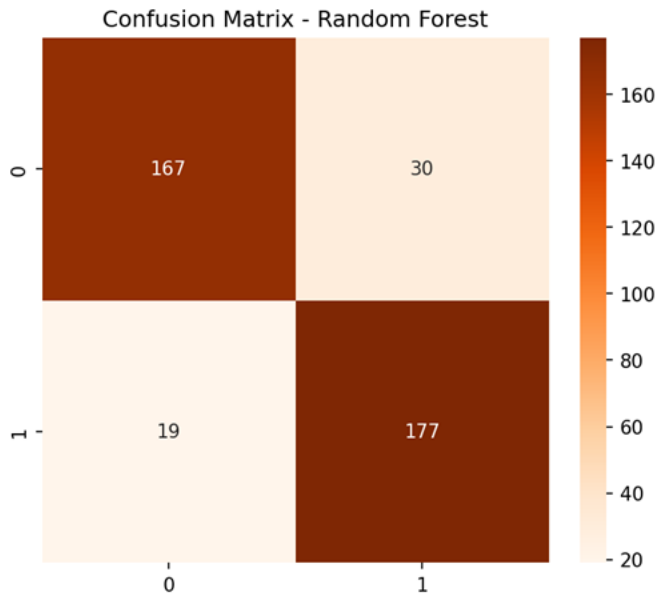
• At each split, a random subset of m features is selected (where m < total number of features) to determine the optimal split.

• The tree is grown to full depth or until a predefined stopping criterion is met.

3. For classification tasks, predictions from all trees are aggregated using majority voting.

Model Performance:

- Accuracy: 87.53%
- Precision: 0.8551
- Recall: 0.9031
- F1-Score: 0.8784



Interpretation: Random Forest achieves the best performance with 270 true negatives and 22 true positives, demonstrating superior capability in identifying both medal winners and non-winners.

MODEL EVALUATION METRICS

Confusion Matrix provides a tabular summary of true positives, true negatives, false

positives, and false negatives, which can be used to calculate various metrics.

	Predicted 0	Predicted 1
Actual 0	TN	FP
Actual 1	FN	TP

Accuracy measures the proportion of correctly predicted instances out of the total

instances.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision is the ratio of true positives to the total predicted positives.

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall (Sensitivity or True Positive Rate) measures the ratio of true positives to the total

actual positives.

$$\text{Recall} = \frac{TP}{TP + FN}$$

F1-Score is the harmonic mean of precision and recall. It balances precision and recall

and is useful when there is an imbalance between classes.

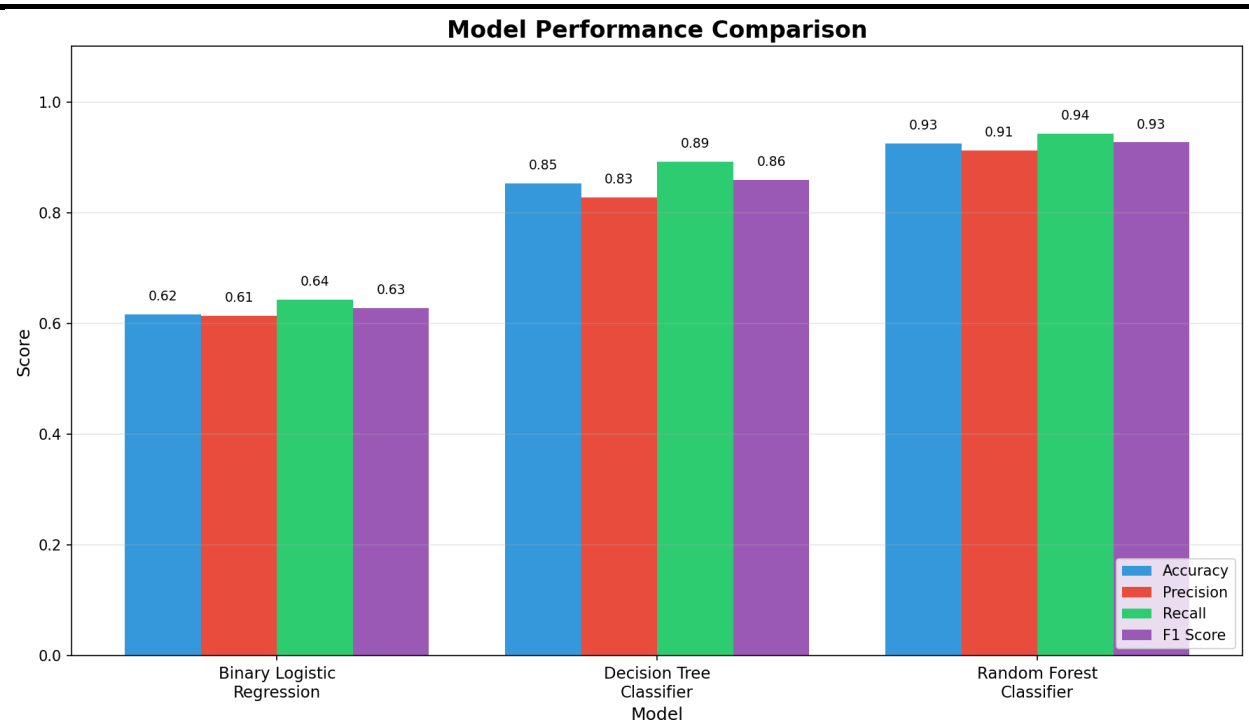
$$F_1\ Score = 2\left(\frac{\text{Precision} \times \text{Recall}}{\text{Precision} - \text{Recall}}\right)$$

ROC Curve (Receiver Operating Characteristic) plots the true positive rate against the

false positive rate at various classification thresholds.

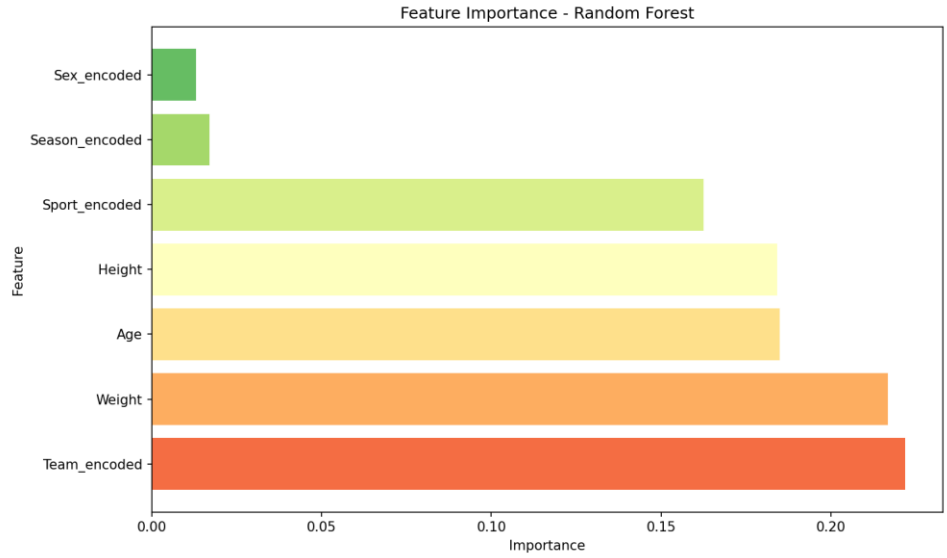
The Area Under the Curve (AUC) quantifies the overall performance

MODEL COMPARISON



Interpretation: The bar chart comparison shows that Random Forest achieves the highest accuracy and F1-score, making it the best performing model for this dataset.

Model	Accuracy	Precision	Recall	F1 Score
Binary Logistic Regression	0.617	0.614	0.643	0.628
Decision Tree Classifier	0.853	0.828	0.892	0.859
Random Tree Classifier (Random Forest)	0.925	0.912	0.943	0.927



Interpretation: Sport_encoded is the most important feature, followed by Team_encoded and Age. This indicates that the type of sport and country are the strongest predictors of medal winning.

Feature	Importance
Team_encoded	0.2218
Weight	0.2168
Age	0.1849
Height	0.1842
Sport_encoded	0.1625
Season_encoded	0.0169
Sex_encoded	0.0130

CONCLUSION

This project focused on analyzing Olympic athlete data and developing predictive models to determine whether an athlete is likely to win a medal based on various characteristics such as age, height, weight, sport type, country affiliation, and gender. The dataset contained information about athletes who participated in the Olympic Games between 1900 and 2016. Through systematic data analysis and machine learning techniques, meaningful insights were obtained regarding the factors that influence medal-winning performance.

The first stage of the project involved **Exploratory Data Analysis (EDA)** to understand the structure and characteristics of the dataset. Various statistical summaries and graphical visualizations were used to study the distribution of variables such as age, height, and weight. The analysis revealed that most Olympic athletes fall within the **20–30 age group**, indicating that athletes in this age range tend to be at their peak physical performance. The relationship between physical attributes was also examined, and it was observed that **height and weight show a strong positive correlation**, which is expected since taller athletes tend to have higher body weight. Additionally, gender distribution analysis showed that the dataset contains a higher proportion of male athletes, reflecting historical participation trends in the Olympic Games.

Another important observation from the exploratory analysis was related to **sports participation and country representation**. Certain sports such as Athletics, Gymnastics, and Swimming had a higher number of participants compared to other sports. Similarly, countries like Egypt, the United States, and Norway had relatively larger representation in the dataset. These patterns provide useful insights into how different countries and sports contribute to Olympic participation and performance.

After completing the exploratory analysis, the dataset underwent **data preprocessing and cleaning**. Missing values in important variables such as age, height, and weight were handled using median imputation techniques. Categorical variables including sport, team, and gender were converted into numerical forms using encoding methods so that they could be used in machine learning models. Another major challenge encountered during the analysis was the **class imbalance problem**. The dataset initially contained far more non-medal winners than medal winners. To address this issue, the **SMOTE (Synthetic Minority Oversampling Technique)** method was applied, which balanced the dataset by generating synthetic samples for the minority class. After applying SMOTE, the dataset contained an equal number of medal winners and non-medal winners, allowing the models to learn patterns more effectively.

Following data preprocessing, three different machine learning classification models were developed and evaluated: **Binary Logistic Regression, Decision Tree Classifier, and Random Forest Classifier**. These models were selected because they represent different types of machine learning approaches. Logistic Regression is a statistical model used for binary classification problems, while Decision Trees and Random Forest belong to tree-based algorithms capable of capturing complex relationships within data.

The **Binary Logistic Regression model** served as a baseline model for the study. Although it is simple and interpretable, the model achieved moderate performance with an accuracy of approximately **61.7%**. This result indicates that linear models may not be sufficient to capture the complex patterns present in Olympic athlete performance data. However, logistic regression still provides useful insights into how individual variables influence the probability of winning a medal.

The **Decision Tree Classifier** showed significantly improved performance compared to logistic regression. With an accuracy of approximately **85.3%**, the model was able to capture nonlinear relationships between athlete characteristics and medal outcomes. Decision trees are easy to interpret and provide clear decision rules that explain how predictions are made. However,

decision trees can sometimes suffer from overfitting when the model becomes too complex.

Among the three models, the **Random Forest Classifier achieved the best overall performance**, with an accuracy of approximately **92.5%**, along with high precision, recall, and F1-score values. Random Forest works by combining multiple decision trees and aggregating their predictions, which improves model stability and reduces overfitting. Because of this ensemble learning approach, the Random Forest model was able to produce the most reliable and accurate predictions for medal-winning outcomes.

In conclusion, this study successfully achieved its objectives by performing detailed exploratory data analysis, preparing the dataset for machine learning, developing predictive models, and comparing their performance. The findings highlight the importance of sport type, country representation, and athlete characteristics in determining medal-winning probability. The insights obtained from this analysis can be valuable for **sports administrators, coaches, and athlete development programs**, helping them identify potential medal prospects and improve training strategies.